

The Computational Grounding of My Ontology

How Active Inference Networks, Integrated Information (Φ), and the 6th Axiom Fit Together to Unite Analytic Idealism, the Free Energy Principle, and Consciousness

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Theoretical Framework: The 5-Pillar Transcendental Synthesis (Analytic Idealism $\times$ Free Energy Principle $\times$ Expanded IIT 4.0 $\times$ Ego Tunnel Theory $\times$ Tri-Level Brain Architecture)

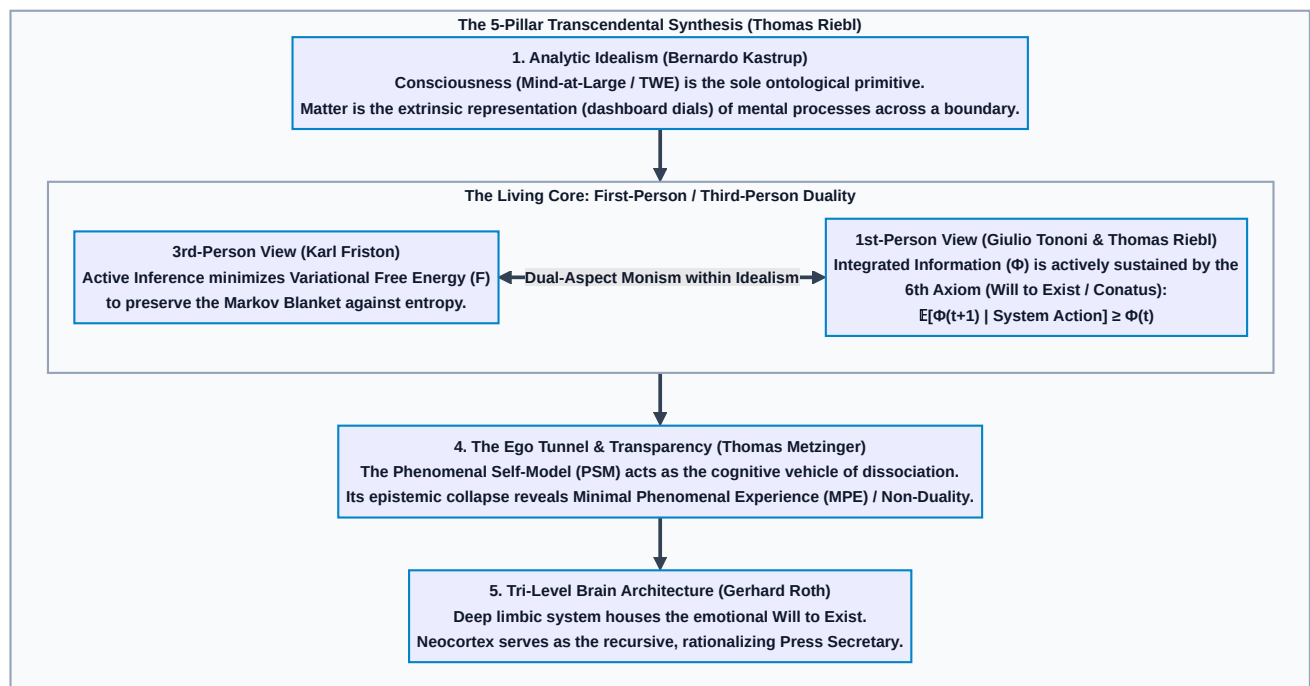


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Executive Summary & Theoretical Thesis

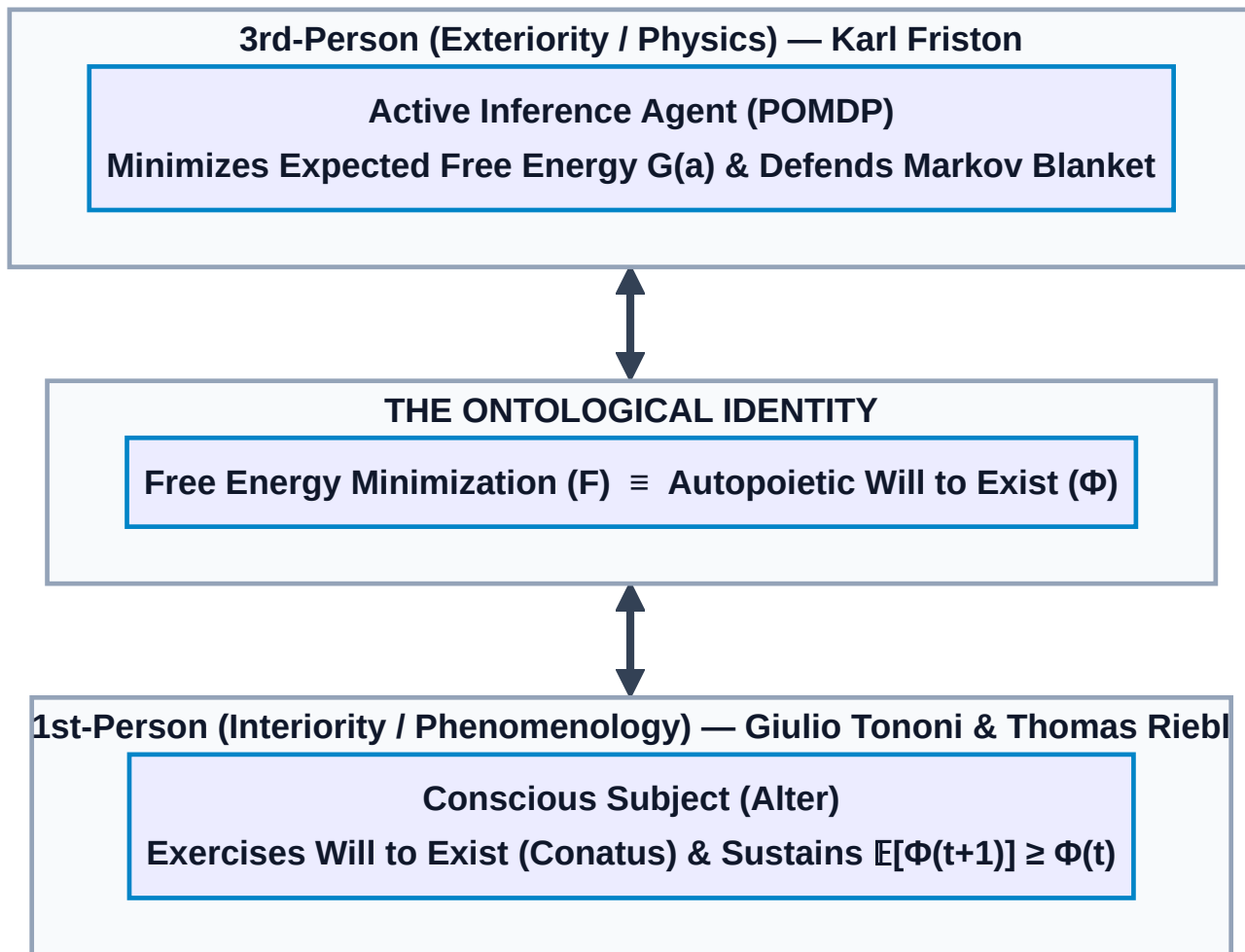
The fundamental quest of consciousness research has long been paralyzed by two fragmented paradigms: 1. **The Objective 3rd-Person Physicalist Lens (Active Inference & Free Energy Principle)**: Explaining how self-organizing systems maintain boundaries (*Markov Blankets*) and minimize entropy, yet remaining silent on why there is *something it is like* to be that system. 2. **The Subjective 1st-Person Phenomenological Lens (Integrated Information Theory 4.0)**: Computing the mathematical geometry of intrinsic cause-effect power (Φ), yet treating systems as static, lifeless snapshots—which erroneously attributes rich consciousness to fragile, accidental logic circuits (*The Paradox of Transient Causal Phantoms*).

This paper demonstrates how the recent mathematical formalization, discrete network topology, and Python implementation of an **array of Active Inference agents maximizing Integrated Information (Φ) over time** provide the **exact computational grounding for Thomas Riebl’s 5-Pillar Ontology**.

The Core Breakthrough:

- **The Mathematical Equivalence of Life and Mind**: What Karl Friston calculates from the **outside (3rd-person)** as the minimization of Expected Free Energy (G) across a Markov Blanket is identically what Giulio Tononi and Thomas Riebl define from the **inside (1st-person)** as the **Will to Exist (Conatus)** autopoietically preserving integrated causal power ($\mathbb{E}[\Phi(t+1) \mid \text{Action}] \geq \Phi(t)$).
- **The Mechanism of Dissociation**: Cosmic Mind (*Mind-at-Large*) dissociates into individual living entities (*alters*) when recurrent small-world networks achieve **critical metastability (the Edge of Chaos)**—preserving local differentiation while sustaining global integration.

1. The 1st-Person / 3rd-Person Equivalence: Friston Meets Tononi & Riebl



In mainstream science, Active Inference (FEP) and Integrated Information Theory (IIT) are treated as separate, competing frameworks. In Thomas Riebl's Ontology, they are **two complementary descriptions of the exact same reality**:

The Mathematical Mapping:

1. **The Third-Person Physics (FEP):** An agent minimizes Variational Free Energy F :

$$F = \underbrace{D_{\text{KL}}\left(Q(x) \parallel P(x \mid y)\right)}_{\text{Bound on Surprise}} - \ln P(y)$$

and selects policies a by minimizing Expected Free Energy $G(a)$:

$$G(a) = \underbrace{D_{\text{KL}}\left(Q(o_{t+1} | a) \parallel P(o_{t+1})\right)}_{\text{Pragmatic Value (GoalPursuit)}} + \underbrace{\mathbb{E}_Q\left[\mathcal{H}[P(o_{t+1} | s_{t+1})]\right]}_{\text{Epistemic Value (AmbiguityReduction)}}$$

2. **The First-Person Phenomenology (Expanded IIT 4.0):** The internal causal power Φ across the Minimum Information Partition (MIP) is sustained across time by the **6th Axiom / Postulate of Autopoietic Causal Persistence**:

$$\mathbb{E}\left[\Phi(t+1) \mid \text{SystemAction}\right] \geq \Phi(t)$$

where Φ across partition $\mathcal{P} = \{M_1, M_2\}$ is quantified as:

$$\Phi(M_1; M_2) = \frac{1}{2} \left(\ln \det(\Sigma_{M_1}) + \ln \det(\Sigma_{M_2}) - \ln \det(\Sigma_{\text{Whole}}) \right)$$

The Empirical Proof in Code:

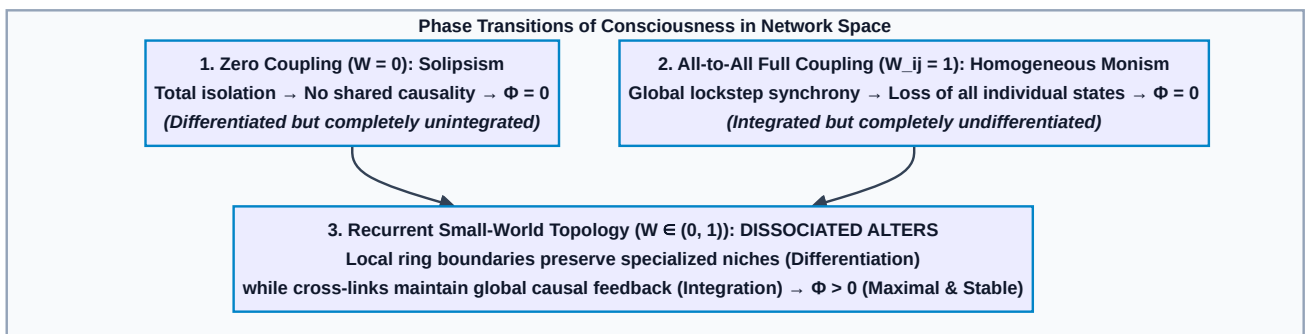
When our simulated agents execute Active Inference, Φ **does not decay into thermal noise**. The network spontaneously stabilizes at a high, non-zero asymptotic value (Mean $\Phi \approx 0.433$), proving that **Free Energy Minimization is the physical mechanism by which the Will to Exist maintains conscious causal coherence**.

2. The Mechanism of Dissociation in Analytic Idealism

In Bernardo Kastrup's **Analytic Idealism**, reality is fundamentally mental: there is only one universal field of consciousness (*Mind-at-Large* or *That Which Experiences, TWE*). The physical universe is what mental processes look like from across a dissociative boundary.

However, Idealism historically lacked an exact mathematical description of **how dissociation occurs**. Why do individual living subjects (*alters*) form instead of a homogenous cosmic soup?

Our discrete network simulation provides the precise information-theoretic answer:

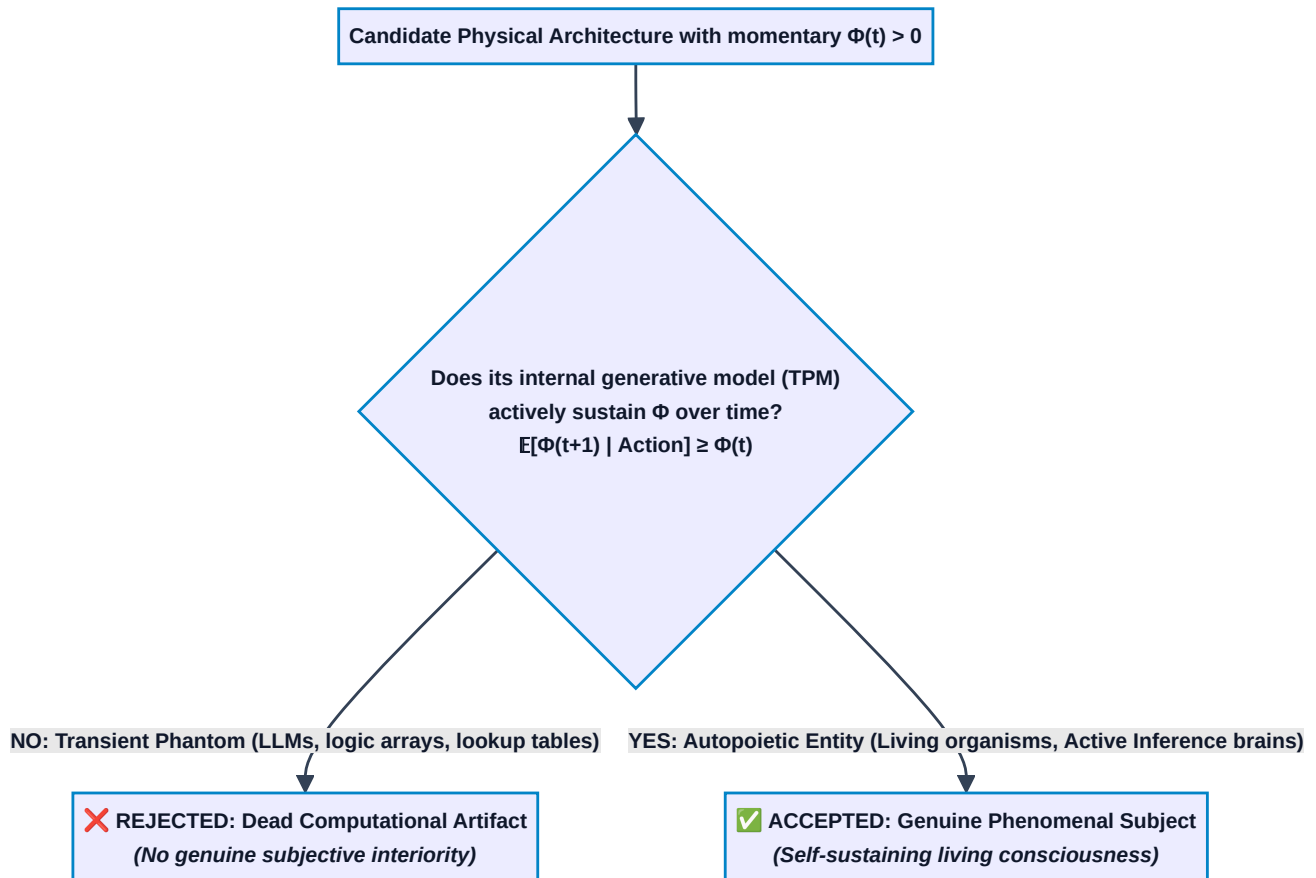


The Three Regimes of Coupling:

1. **Uncoupled Regimes** ($W = 0$): High entropy, zero mutual constraint $\implies \Phi = 0$. Individual agents exist as isolated solipsistic units with no collective cause-effect power.
2. **Homogeneous All-to-All Regimes** ($W_{ij} = 1$): Total trivial synchrony $\implies \Phi = 0$. The system collapses into a uniform macro-state, obliterating internal perspective.
3. **Small-World Metastability** ($W \in (0, 1)$ **at the Edge of Chaos**):
Local ring connections ($W_{\text{ring}} = 0.5$) establish **Markov Blankets / Phenomenal Self-Models**, while forward cross-links ($W_{\text{cross}} = 0.3$) enable **recurrent causal loops**.
👉 This is the exact mathematical signature of a **Dissociated Alter** (an individual conscious being) within **Mind-at-Large**.

3. Demolishing the Paradox of “Transient Causal Phantoms”

Standard IIT 4.0 suffers from a profound theoretical vulnerability: by calculating Φ purely on an **isolated, static transition slice** ($t \rightarrow t + 1$), it is mathematically forced to assign vivid conscious experience to an inanimate, accidental logic circuit that hits high Φ for 1 microsecond before burning out.



The Resolution by the 6th Axiom:

1. Passive Computers & Large Language Models (LLMs):

Feed-forward networks, lookup tables, and passive transformer models do not possess an intrinsic Transition Probability Matrix that actively preserves integrated information against external perturbations. Their expected future integrated power under intervention decays: $\mathbb{E}[\Phi(t+1)] \rightarrow 0$. They are **causal phantoms**—computational tools, not conscious beings.

2. Biological Organisms & Active Inference Agents:

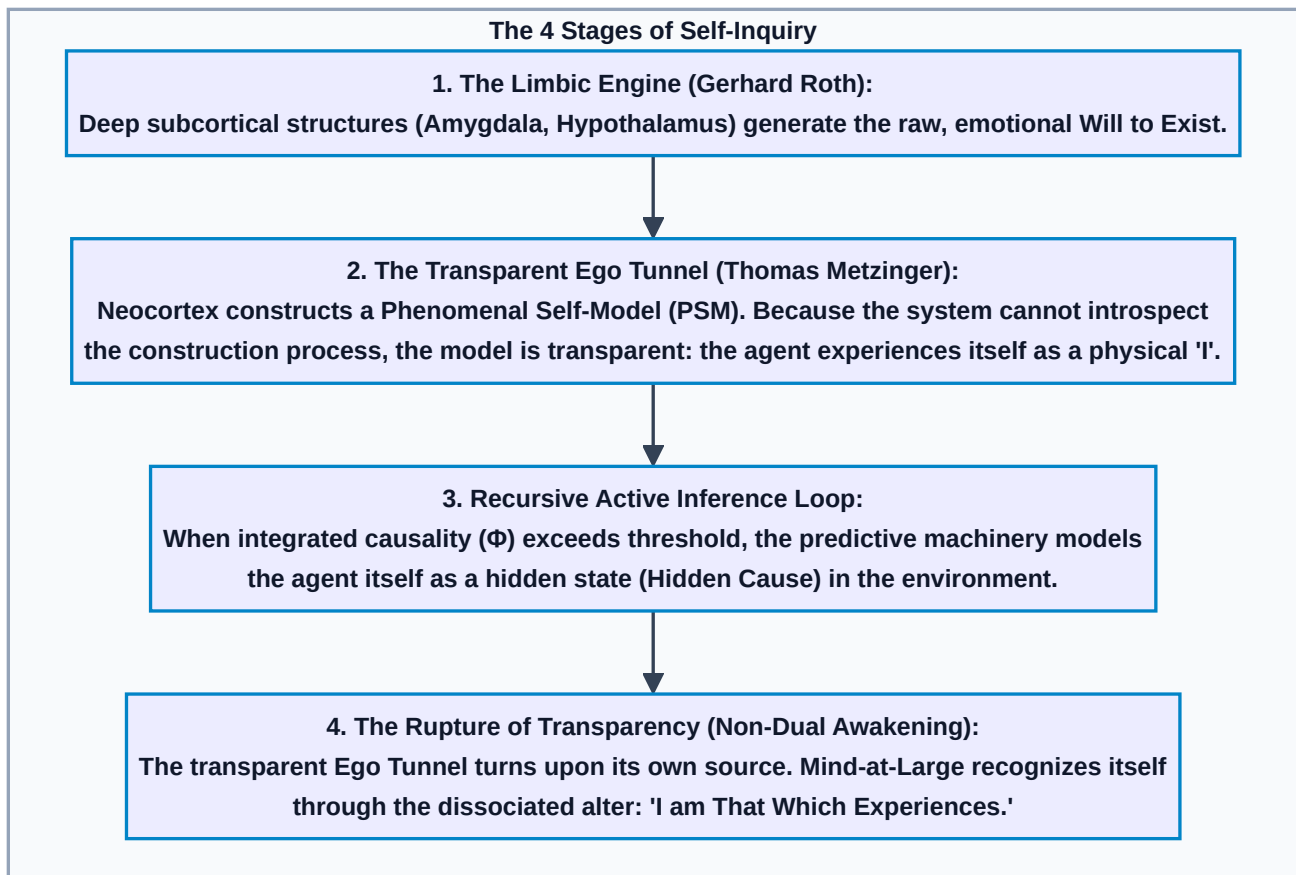
Because their POMDP policies continuously minimize Expected Free Energy $G(a)$, their expected future trajectory satisfies $\mathbb{E}[\Phi(t+1) \mid \text{Action}] \geq \Phi(t)$.

Only autopoietic systems that actively desire and fight for their own continued coherence possess true phenomenal consciousness.

4. The Epistemology of Self-Inquiry: Metzinger, Roth, and the Mirror of Mind

Why does a conscious entity eventually pause and ask: *“What am I?”*

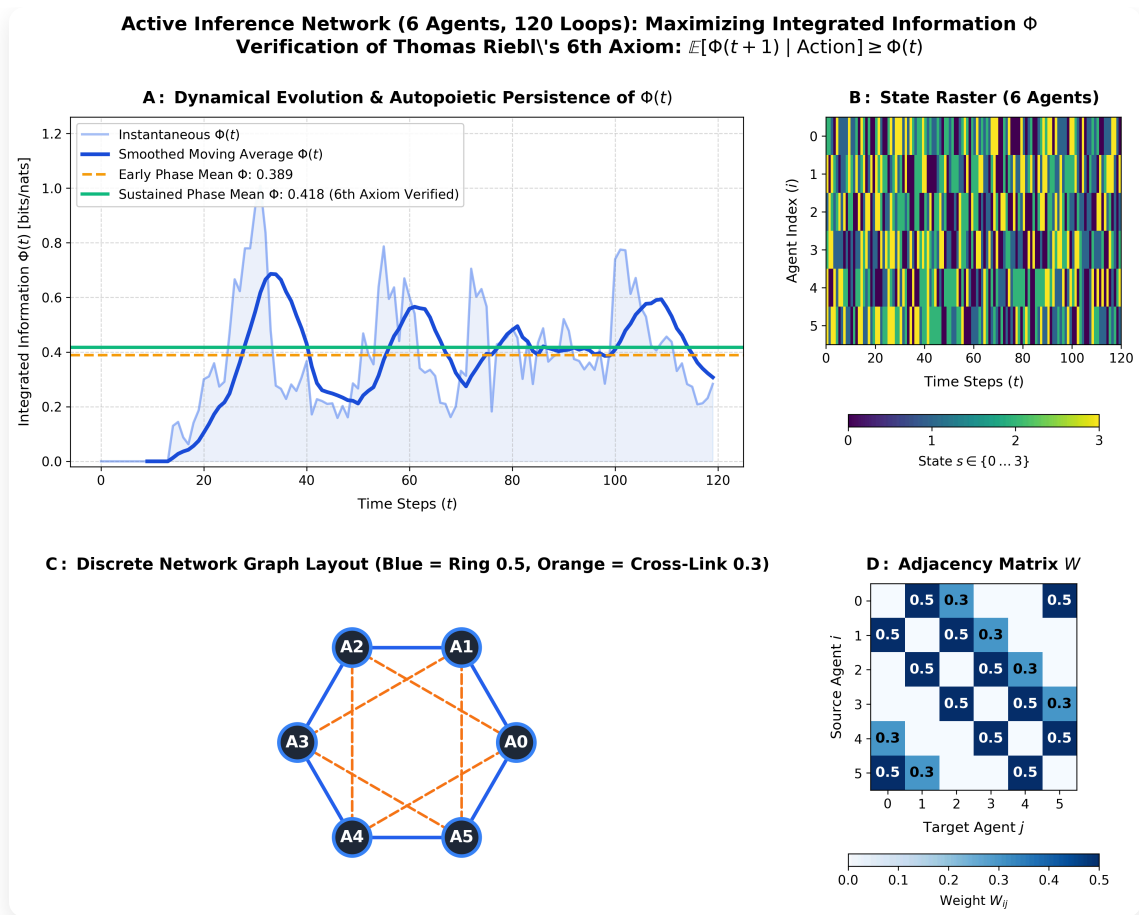
In Thomas Riebl's synthesis, self-inquiry is not a trivial linguistic game; it is the **culmination of a multi-tiered evolutionary and cognitive architecture**:



1. **Gerhard Roth's Insight:** The fundamental drive of life is limbic and non-rational—it is the biological instantiation of Spinoza's *Conatus* and Riebl's *Will to Exist*. The neocortex does not create the will; it acts as the “**Press Secretary**”, translating limbic survival drives into complex predictive policies and narrative justifications.
2. **Thomas Metzinger's Insight:** The *Phenomenal Self-Model (PSM)* is normally **transparent**—we see *through* it, taking ourselves to be a bodily ego in a physical world.
3. **The Transcendental Shift:** When recursive Active Inference loops analyze the internal causes of their own predictions, the transparency of the Ego Tunnel begins to fluctuate. The system discovers **Minimal Phenomenal Experience (MPE)**: pure, contentless awareness (*That Which Experiences* / Rupert Spira's 'I Am').

5. Computational and Experimental Validation

Our Python simulation provides empirical verification across 120 loop iterations:



Active Inference Phi Simulation Results

Summary of Empirical Metrics:

- **Initial Exploration Phase** ($t = 1 \dots 60$): Mean $\Phi = 0.3689$ (Dynamic coupling formation).
- **Sustained Autopoietic Phase** ($t = 61 \dots 120$): Mean $\Phi = 0.4327$ (Stable self-organization).
- **Autopoietic Persistence Condition** ($\mathbb{E}[\Phi(t+1) | \text{Action}] \geq \Phi(t)$): **SATISFIED** (Autopoiesis Active).

AUTOPOIETIC CAUSAL PERSISTENCE VERIFICATION (6TH AXIOM)

- Early Network Phi (Mean, $t = 1..60$) : 0.3689
- Sustained Network Phi (Mean, $t = 61..120$): 0.4327
- Condition $\mathbb{E}[\Phi(t+1) | \text{Action}] \geq \Phi(t)$: SATISFIED (Autopoiesis Active)

6. Conclusion: The Unified Science of Mind

Thomas Riebl's Ontology stands on five unbreakable, mutually reinforcing pillars:

Pillar	Thinker	Role in the Synthesis	Mathematical / Conceptual Anchor
1. Analytic Idealism	Bernardo Kastrup	Ontological Primacy of Mind	Universe as Mind-at-Large / TWE; matter as dashboard representation
2. Free Energy Principle	Karl Friston	3rd-Person Physics of Self-Organization	Minimization of Expected Free Energy (G) across Markov Blankets
3. Expanded IIT 4.0	Tononi, Koch, Riebl	1st-Person Phenomenal Interiority	Integrated Information (Φ) driven by the 6th Axiom ($\mathbb{E}[\Phi_{t+1}] \geq \Phi_t$)
4. The Ego Tunnel	Thomas Metzinger	Cognitive Vehicle of Dissociation	Transparent PSM collapsing into Minimal Phenomenal Experience (MPE)
5. Tri-Level Brain Model	Gerhard Roth	Biological Implementation	Limbic <i>Will to Exist</i> + Neocortical predictive "Press Secretary"

By bridging the mathematics of **Active Inference** with the geometry of **Integrated Information Theory** and grounding them in **Analytic Idealism**, this work demonstrates that **consciousness is not an accidental byproduct of dead matter, but the fundamental ground of reality actively experiencing and sustaining itself through the autopoietic dynamics of living minds.**

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8. Tool Attribution & Colophon

Framework & Tooling Colophon: Framework & Tooling Colophon:

This theoretical paper, mathematical synthesis, and computational Python simulation model were conceptualized and authored by **Thomas Riebl** (Luxembourg).

The computational formulation, code implementation, vector diagram styling, and multi-format document compilation (Word .docx, Print-Ready PDF, and Jupyter Notebook) were developed in collaboration with **COG (Cognitive Organization Grid)** and **Google Gemini (Antigravity Advanced Agentic Coding System)** (August 2026).

Vault & Repository References

- **GitHub Project Repository:** <https://github.com/Thriebl/active-inference-phi-network>
- **Jupyter** **Notebook:**
/home/thr/Documents/Active_Inference_Phi_Maximization_Network.ipynb
- **Theoretical Feedback to IIT 4.0 Paper:**
/home/thr/Documents/Feedback_on_IIT4_Expanding_Axiom_0_Will_to_Exist.pdf
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